
Software Requirements Specification

for

MatchMyHome

Version 1.0.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Group #2	May 18, 2026	Initial draft compiling system overview, scoping boundaries, and external interface configurations	1.0.0
Group #2	May 27, 2026	Added stakeholder registry and compiled formal stakeholder interview Q&A sessions.	1.1.0
Group #2	June 4, 2026	Added functional and non-functional requirements	1.2.0
Group #2	June 11, 2026	Added use case table	1.3.0
Group #2	June 18, 2026	Added high priority, textual detailed use case description and use case diagram.	1.4.0
Group #2	July 9, 2026	Added class diagram and CRC cards	1.5.0
Group #2	July 16, 2026	Updated class diagram with signature methods and their respective parameters and visibilities	1.6.0
Group #2	July 23, 2026	Added state diagrams for two entity classes	1.7.0
Group #2	July 30, 2026	Added sequence diagram for formal use case description	1.8.0
Group #2	August 6, 2026	Updated class diagram with "Party Analysis Pattern"	1.9.0
Group #2	August 13, 2026	Finalized document	2.0.0

1. Introduction

1.1 Purpose

This document specifies the comprehensive software requirements for version 1.0.0 of the **MatchMyHome** platform. **MatchMyHome** is a self-contained, cross-platform digital marketplace and matching engine designed to facilitate secure, intergenerational co-living arrangements. This SRS governs the entire ecosystem, including the consumer web portal, native mobile companion applications, and the centralized administrative control panel.

1.2 Document Conventions

1.2.1 *Typography and Formatting*

This document utilizes a strict visual hierarchy optimized for engineering and compliance audits:

- **Primary Font Family:** Arial (Sans-serif) for all digital reading and cross-platform asset rendering.
- **Document Title:** 24pt, Bold, Right-aligned.
- **Heading 1 (Main Sections):** 18pt, Bold, Left-aligned (e.g., **1. Introduction**).
- **Heading 2 (Sub-sections):** 14pt, Bold, Left-aligned (e.g., **1.1 Purpose**).
- **Heading 3 (Sub-sub-sections):** 12pt, Bold, Italic, Left-aligned (e.g., **1.2.1 *Typography***).
- **Body Text:** 12pt, Regular, Left-aligned, with 1.5 line spacing.

1.2.2 *Requirement Identifier (ID) Convention*

Every discrete requirement within this document is allocated a unique, traceable, and indexable alphanumeric string formatted as follows:

[System Component]-[Requirement Type]-[Three-Digit Serial Number]

- **System Component:** **APP** (Core Platform / Shared Logic), **WEB** (Web Application), **MOB** (Mobile Application), **ADM** (Admin Portal).
- **Requirement Type:** **FR** (Functional Requirement), **NFR** (Non-Functional Requirement), **BR** (Business Rule), **UI** (User Interface Requirement).
- **Example:** **APP-FR-042** represents the 42nd core Functional Requirement of the platform.

1.2.3 Language and Linguistic Conventions

- **Shall:** Indicates a mandatory, legally binding requirement that must be built and verified.
- **Should:** Indicates a high-priority, non-mandatory feature that is highly desirable.
- **May:** Indicates an optional, auxiliary feature that can be deferred to future release cycles.

1.2.4 Visual Captions and Numbering

- **Figures and Diagrams:** Sequentially numbered by section (e.g., *Figure 2.1: System Context Diagram*). Captions appear directly **below** the asset.
- **Tables:** Sequentially numbered by section (e.g., *Table 3.4: Data Dictionary*). Captions appear directly **above** the table.

Table 1.1: Acronyms and Abbreviations

Acronym	Expanded Term	Context / Definition within MatchMyHome
AA	Level AA (WCAG)	The mid-level accessibility conformance tier required by global and Canadian legal frameworks for web design
ADM	Admin Portal	System component designation used in requirement identifiers for internal administration interfaces

Acronym	Expanded Term	Context / Definition within MatchMyHome
API	Application Programming Interface	A set of protocols that allows MatchMyHome to communicate with external services like Stripe or Google Maps
APP	Application / Core Platform	System component designation used in requirement identifiers for shared logic or platform-wide rules
AWS	Amazon Web Services	A secure, commercial cloud computing platform used to host the containerized server infrastructure
BR	Business Rule	A requirement category designation used in identifiers to specify operational or legal constraints
FCM	Firebase Cloud Messaging	A cross-platform messaging solution used to securely dispatch real-time push notifications to mobile devices
FR	Functional Requirement	A requirement category designation defining specific technical capabilities or system actions that must be built
GPS	Global Positioning System	Onboard mobile hardware sensor used to determine geographical coordinates for location-based matching
GUI	Graphical User Interface	The visual interactive display elements through which users engage with the software
HTTP / HTTPS	Hypertext Transfer Protocol (Secure)	The encrypted communication network protocol used to securely transfer data between client applications and servers
ID	Identifier	A unique, traceable alphanumeric tag assigned to specific system items and requirements

Acronym	Expanded Term	Context / Definition within MatchMyHome
iOS	iPhone Operating System	The native mobile operating system developed by Apple Inc. for iPhones and iPads
JSON	JavaScript Object Notation	A lightweight, human-readable structured data format used for API payloads and database serialization
MOB	Mobile Application	System component designation used in requirement identifiers for native iOS and Android companion applications
NFR	Non-Functional Requirement	A requirement category designation that defines system quality attributes like performance, security, and usability
NoSQL	Non-SQL / Not Only SQL	A non-tabular database design optimized for highly flexible document storage, used by MongoDB Atlas
PII	Personally Identifiable Information	Sensitive user data (e.g., legal names, birthdates, home addresses) protected by strict privacy laws
PIPEDA	Personal Information Protection and Electronic Documents Act	The Canadian federal privacy law governing how private-sector organizations collect, use, and disclose personal data
QA	Quality Assurance	The engineering phase focused on systematically verifying that software features match specification requirements
REST	Representational State Transfer	An architectural software style for distributed systems that uses stateless, standard HTTP methods
SRS	Software Requirements Specification	The formal technical blueprint document specifying the system's complete operational and engineering parameters

Acronym	Expanded Term	Context / Definition within MatchMyHome
TBD	To Be Determined	A temporary placeholder indicating that a specific design choice or technical detail has not yet been finalized
TLS	Transport Layer Security	A cryptographic protocol designed to provide end-to-end communication security over networks (currently v1.3)
UI	User Interface	A requirement category designation used in identifiers to mark specific user layout rules or screens
UX	User Experience	The comprehensive design discipline focused on how users logically navigate and feel while interacting with an application
WCAG	Web Content Accessibility Guidelines	The international framework of technical standards defining how to make digital content accessible to users with disabilities or seniors
WEB	Web Application	System component designation used in requirement identifiers for the desktop-and-tablet web portal interface

1.3 Intended Audience and Reading Suggestions

This document serves as the authoritative blueprint for the MatchMyHome project ecosystem. It is organized to serve diverse reader profiles across the project team:

- **System Architects & Backend Developers (e.g., Michael Rodriguez):** Focus heavily on Sections 2.1, 2.2, 2.5, and Section 3 to map out schemas, REST API integrations, and third-party background verification pipelines.
- **System Administrators (e.g., Daniel Park):** Focus on Section 2.3 (User Classes), Section 3.1 (User Interfaces), and Section 4 (System Features) to understand platform moderation controls, flagged profile review workflows, and administrative dashboard requirements.

- **Senior Homeowners / Hosts (e.g., Margaret Liu):** Prioritize Section 1.4 (Product Scope) and Section 2.3 (User Classes) to understand platform benefits, safety measures, and the accessibility accommodations designed specifically for the senior demographic.
- **Project Managers & Scrum Masters (e.g., Sarah Chen):** Consult Sections 1.4, 2.6, and 2.7 to plan sprint milestones, split product epics, and mitigate environmental risk factors.
- **Tenants / Students & Young Professionals (e.g., Jason Nguyen):** Review Section 2.2 (Product Functions), Section 3.1 (User Interfaces), and Section 4 (System Features) to understand the matching engine, search and filtering capabilities, and messaging tools available to tenant users.
- **Marketing, Investors, & User Advocacy Representatives (e.g., David Okonkwo):** Review Section 1.4 and Section 2.3 to maintain proper demographic messaging alignment, project scope validation, and high-level return on investment (ROI) expectations.

1.4 Product Scope

MatchMyHome bridges two pressing socio-economic challenges in Canada: the profound social isolation experienced by seniors residing alone in underutilized houses, and the severe lack of safe, affordable urban housing available to students and young professionals.

1.4.1 Objectives and Benefits

- **Lifestyle-Based Matching Engine:** Connects users dynamically using an algorithmic compatibility calculation based on lifestyle variables (sleep habits, pet preferences, noise thresholds, and cleanliness).
- **Intergenerational Support:** Enables mutually beneficial arrangements, helping seniors safely age-in-place via light domestic help and companionship while offering youth structured, affordable housing.
- **Security Baseline:** Lowers trust barriers through integrated background screening and automated framework agreements.

1.4.2 Boundaries and Exclusions

The system acts strictly as an administrative facilitator, matching engine, and agreement registrar. It does not manage physical property maintenance, act as a licensed real estate brokerage, provide property management services, or physically enforce house rule compliance.

1.5 References

The following sources were consulted during the research and development of the **MatchMyHome** platform concept.

[R1] Senior Social Isolation in Canada - Government of Canada Government of Canada. (2015). Social isolation of seniors - Volume I: Understanding the issue and finding solutions. Employment and Social Development Canada.
<https://www.canada.ca/en/employment-social-development/corporate/seniors-forum-federal-provincial-territorial/social-isolation-toolkit-vol1.html>

[R2] Senior Loneliness Statistics - Statistics Canada Statistics Canada. (2023). A look at loneliness among seniors.
<https://www.statcan.gc.ca/o1/en/plus/4881-look-loneliness-among-seniors>

[R3] Senior Loneliness Research Summary - Government of Canada

Government of Canada. (2026). Research summary: Seniors' Loneliness in Canada.

<https://www.canada.ca/en/employment-social-development/corporate/reports/research/senior-loneliness-canada.html>

[R4] Social Isolation Epidemic Among Older Canadians - National Institute on Ageing

National Institute on Ageing. (2023). Understanding the factors driving the epidemic of social isolation and loneliness among older Canadians.

<https://www.niageing.ca/loneliness23>

[R5] Youth Housing Crisis in Canada - Youthful Cities

Youthful Cities. (2024). Priced out: The state of youth housing in Canada's urban centres.

<https://www.youthfulcities.com/blog/2024/12/12/priced-out-the-state-of-youth-housing-in-canadas-urban-centres/>

[R6] Affordable Housing Survey - Habitat for Humanity Canada

Habitat for Humanity Canada. (2024). 2024 survey highlights the ripple effect of the housing crisis.

<https://habitat.ca/en/news/2024housingurvey>

[R7] Housing Challenges Among Young Adults - Statistics Canada

Statistics Canada. (2024). Housing challenges related to affordability, adequacy, condition and discrimination.

<https://www150.statcan.gc.ca/n1/daily-quotidien/241119/dq241119b-eng.htm>

[R8] Silvernest / HomeShare Online - Comparable Platform

HomeShare Online (formerly Silvernest). Senior homesharing and roommate finder platform.

<https://www.silvernest.com>

2. Overall Description

2.1 Product Perspective

MatchMyHome is a new, self-contained software product with no predecessor system. It addresses a gap in the Canadian market where no dedicated platform currently pairs seniors with underutilized home space with students or young professionals seeking affordable housing. While standalone, **MatchMyHome** will interface with third-party identity verification services, a criminal background check API, a secure payment gateway, and the Google Maps API for location-based features.

2.2 Product Functions

The platform executes the following primary system operations:

- **Profile Identity Lifecycle:** Onboarding, profile curation, and custom preference entry for both hosts and tenants.
- **Biometric & Background Screening:** Secure payload routing to verification systems to clear users before unlocking interactions.
- **Geospatial Listing Discovery:** Radius-based search mapping and boundary visualizations via map layers.
- **Lifestyle Match Score Evaluation:** Context-aware data correlation processing user habits into prioritized compatibility indexes.
- **Real-Time Messaged Communication:** Secure, in-app text routing for user discovery without revealing raw personal metadata.
- **Chore & Living Agreement Formulator:** Structural tools for outlining specific household service contributions in exchange for rental discounts.
- **Trust Ledger & Reviews:** Peer-to-peer feedback systems documenting completed arrangements.

2.3 User Classes and Characteristics

The **MatchMyHome** platform segregates users into four distinct operational classifications. These classes are differentiated by their technical literacy, frequency of use, and platform privilege levels.

2.3.1 Seniors (Homeowners / Hosts) - Primary Critical Class

This is the platform's most vital and risk-sensitive demographic. Satisfying their specific safety and accessibility requirements is paramount to the product's market viability.

- **Description:** Adults aged 55+ possessing underutilized property space.
- **Technical Expertise:** Low to varying levels of technical literacy.
- **Frequency of Use:** Intermittent (primarily during onboarding, listing creation, and initial matching).
- **Key Functions Used:** Property listing management, profile creation, and communication portals.
- **Design & Safety Needs:** Highly accessible user flows, large visual layouts, plain language, and explicit identity safety gates.

2.3.2 Tenants (Students / Young Professionals) - High Importance Class

This demographic drives the demand side of the marketplace application and expects a modern, seamless digital experience.

- **Description:** Digital-native adults aged 18–35 seeking affordable rent.
- **Technical Expertise:** Expert proficiency with modern web and mobile platforms.
- **Frequency of Use:** High / Daily (until a successful match is secured).
- **Key Functions Used:** Advanced search and filtering, mapping interfaces, tenant profile building, and digital messaging.
- **Design & Safety Needs:** Transaction speed, search engine clarity, and rapid interface fluidness.

2.3.3 System Administrators - Essential Operational Class

The internal corporate operations team responsible for maintaining platform integrity, safety, and daily matching health.

- **Description:** Internal operational staff with expert software knowledge utilizing desktop environments.
- **Technical Expertise:** Expert.
- **Frequency of Use:** Continuous / Daily.
- **Key Functions Used:** Safety metric monitoring dashboards, flagged profile review queues, operational contract frameworks, and billing exception audits.
- **Security & Privilege Level:** Administrative. Requires elevated read/write permissions to user databases and platform moderation controls.

2.3.4 Technical Creators & Maintainers - Internal Engineering Class

The backend operational stakeholders from the project register who use the system's development infrastructure to deploy updates, maintain security, and optimize user experience.

- **Description:** The internal project team responsible for the software's structural and operational lifecycles.
- **Stakeholders Included:**
 - **Sarah Chen (Project Manager):** Uses project tracking modules and requirement verification analytics.
 - **Michael Rodriguez (Developer):** Uses code repositories, API integration environments, and backend database architecture.
- **Technical Expertise:** Advanced / Expert.
- **Frequency of Use:** Continuous / Throughout the project lifecycle.
- **Security & Privilege Level:** Deep Infrastructure Access (Write permissions for codebase, staging servers, and security protocols).

2.4 Operating Environment

The **MatchMyHome** architecture shall execute reliably within the following runtime constraints:

- **Cloud Infrastructure:** Hosted on AWS or Azure using containerized clustering.
- **Database Management Layer:** MongoDB Atlas (v7.0+) running NoSQL configurations.
- **Backend Server Runtime:** Node.js/Express configured as an asynchronous RESTful API layer processing `application/json` requests, coupled with a stateful WebSocket server architecture (`ws://` / `wss://`) to handle real-time chat infrastructure.
- **Web Interfaces:** React engine web applications built to support Google Chrome (v110+), Mozilla Firefox (v110+), and Apple Safari (v16+).
- **Mobile Interfaces:** Native application runtimes built via React Native supporting Apple iOS (v15+) and Google Android (v10+).

2.5 Design and Implementation Constraints

- **Regulatory Compliance:** The system must enforce data schemas that strictly conform to Canada's *Personal Information Protection and Electronic Documents Act* (PIPEDA) and provincial consumer protection/human rights laws regarding housing.
- **Accessibility Mandate:** To ensure equitable usage by the Senior homeowner class, all user-facing web text views must conform to Web Content Accessibility Guidelines (WCAG) 2.1 Level AA parameters.
- **State Management & Architecture:** System components must interact exclusively over secure HTTP channels (TLS 1.3) via a stateless REST API layout.

2.6 User Documentation

The product release plan must include the delivery of the following customer support assets:

- **Senior In-App Guided Onboarding:** Interactive walk-through wizards embedded inside the web portal.
- **Digital Knowledge Base:** A searchable online repository containing step-by-step visual documentation, privacy descriptions, and safety standard definitions.
- **System Administrator Training Manual:** A technical guide detailing review mechanics, dispute tickets, and escalation patterns.

2.7 Assumptions and Dependencies

- **Network Availability:** Assumes end users have baseline access to stable data networks or internet connectivity.
- **Geographic Focus:** Operational parameters assume users reside natively within Canadian territories and provide Canadian documentation.
- **API Continuity:** Assumes uninterrupted operational runtime, formatting compatibility, and business continuity of Google Maps API, Stripe Payment Gateway, and Certn/Sterling background screening networks.
- **Data Integrity:** Assumes information inputs submitted by users during verification registration are legally accurate.

3. External Interface Requirements

3.1 User Interfaces

3.1.1 General UI Design Standards

- **WEB-UI-001:** The Web application shall implement responsive breakpoints for standard Desktop (1920x1080) and Tablet (1024x768) dimensions using an Arial font core.

- **WEB-UI-002:** The web portal interface shall provide an accessible interface toggle conforming to WCAG 2.1 AA standards, supporting font sizing expansion up to 200% and a high-contrast color palette layout ($> 4.5:1$ contrast ratio).
- **MOB-UI-001:** The Mobile interface shall leverage native device screen ratios on iOS and Android platforms, displaying adaptive grid formats using React Native rendering frameworks.

3.1.2 Required System Interface Screens

The platform must implement custom user interfaces for the following views:

Table 3.1: Required System Interface Screens and Platform Support

View Name	Web Support	Mobile Support	Admin Support	Target Release Phase	Functional Scope
Authentication Screen	Yes	Yes	Yes	Phase 1 (Web/Admin) Phase 2 (Mobile)	Login, Registration, and Multi-Factor Auth
User Dashboard	Yes	Yes	Yes	Phase 1 (Web/Admin) Phase 2 (Mobile)	Context-aware entry portal personalized by user type
Account Profile Wizard	Yes	Yes	No	Phase 1 (Web) Phase 2 (Mobile)	Curation page for lifestyle attributes and photos

View Name	Web Support	Mobile Support	Admin Support	Target Release Phase	Functional Scope
Map Discovery Board	Yes	Yes	No	Phase 1 (Web) Phase 2 (Mobile)	Radius search map showing location-filtered matches
Matching Matrix Panel	Yes	Yes	No	Phase 1 (Web) Phase 2 (Mobile)	Interactive view comparing habit compatibility indexes
Secure Message Room	Yes	Yes	No	Phase 1 (Web) Phase 2 (Mobile)	Encrypted communication screen with reporting buttons
Agreement Workspace	Yes	Yes	No	Phase 1 (Static Document Template) Phase 2 (Interactive Workspace Canvas)	In-app workspace for structuring household chores
Admin Control Center	No	No	Yes	Phase 1 (Admin)	Central console for identity validation overrides

View Name	Web Support	Mobile Support	Admin Support	Target Release Phase	Functional Scope
Contract Ledger Board	No	No	Yes	Phase 1 (Admin)	Administrative tracking panel for active agreements

3.2 Hardware Interfaces

The software does not control physical hardware units directly, but interacts with device hardware subsystems through host operating system wrappers:

- **APP-HI-001 (Camera Subsystem):** Client applications shall interface with mobile and web camera modules to execute profile photo capture and identity document image scanning.
- **MOB-HI-001 (Location Sensors):** Mobile application packages shall interface with onboard GPS units to capture relative proximity positions for localized housing radius calculations.
- **MOB-HI-002 (Biometric Hardware):** Mobile deployment packages shall interface with secure biometric platforms (Apple FaceID/TouchID and Android Biometric Prompt API) to execute local user authentication.

3.3 Software Interfaces

MatchMyHome will interface with the following third-party software systems:

Google Maps API (v3.55+): Used for location-based home listing, search radius filtering, and displaying nearby matches on a map. Documentation:

<https://developers.google.com/maps/documentation>

Stripe Payment Gateway (API v2024+): Used to facilitate secure rent transactions between tenants and senior homeowners. Documentation: <https://stripe.com/docs/api>

Certn or Sterling Background Check API: Used to conduct criminal record and identity verification checks on all registered users prior to matching. Documentation: <https://docs.certn.co>

Firebase Cloud Messaging (FCM): Used to deliver push notifications to mobile users for new matches, messages, and agreement updates. Documentation: <https://firebase.google.com/docs/cloud-messaging>

MongoDB Atlas (v7.0+): Cloud-hosted NoSQL database used to store and retrieve user profiles, match data, and agreement records. Documentation: <https://www.mongodb.com/docs/atlas>

3.4 Communications Interfaces

- **APP-CI-001 (Transport Layer Protection):** Every data network transaction executed within the application environment shall enforce HTTPS connection rules wrapped inside TLS 1.3 encryption mechanisms.
- **APP-CI-002 (Data Format Serialization):** All communication between structural database layers, internal service systems, and target consumer clients shall be structured using strict JSON formats over REST API patterns.
- **APP-CI-003 (Asynchronous Event Alerts):** Real-time web messaging pipelines shall execute via WebSocket communication connections, falling back to HTTPS long-polling patterns if network environments degrade.

3.5 Use Cases

3.5.1 Table: Sub-systems/modules 1: User Profile & Verification Management

User Case Name	Related Requirements ID	Actor (s)	Brief Description
Create-Profile	WEB-FR-001	Senior, Tenant	The actor starts the onboarding process and enters personal information, housing preferences, lifestyle details, and profile information. The system validates the entered data and stores the profile information in the database. Upon successful completion, the system creates a profile that can participate in the matching process.
Upload-Documents	APP-FR-002	Senior, Tenant	The actor uploads the required identification documents and profile photos through the verification interface. The system validates the file format and securely stores the submitted documents. The uploaded information is then made available for identity verification and background screening processes.
Verify-Identity	APP-FR-003	Senior, Tenant	The actor submits identity verification information and completes the required biometric verification steps. The system securely transmits the verification data to the identity verification service and evaluates the results. Upon successful verification, the system updates the actor's profile status to "verified".

User Case Name	Related Requirements ID	Actor (s)	Brief Description
Complete-BackgroundCheck	APP-FR-004	Senior, Tenant	The actor provides the information required for background screening and consents to the verification process. The system securely packages and transmits the data to the Certn/Sterling screening service. After the screening process is completed, the system records the results and updates the actor's account status accordingly.
Override-Validation	ADM-FR-011	System Administrator	The administrator reviews a profile that has been restricted by automated validation rules. The system presents the reason for the restriction along with supporting evidence. Based on the review, the administrator can approve the account, maintain the restriction, or reject the profile.

3.5.2 Table: Sub-systems / modules 2: Matching & Communication

User Case Name	Related Requirements ID	Actor (s)	Brief Description
Search-Listings	APP-FR-005	Tenant	The actor enters location preferences and search criteria to find available housing opportunities. The system performs geospatial searches using the specified radius and displays matching listings on an interactive map. The actor can review the search results and select listings for further evaluation.

User Case Name	Related Requirements ID	Actor (s)	Brief Description
Evaluate-Compatibility	APP-FR-006	Senior, Tenant	The actor reviews compatibility information for a potential match. The system analyzes lifestyle preferences and matching criteria and calculates a compatibility score. The actor can use this information to determine whether to proceed with communication or housing arrangements.
Send-Message	APP-FR-007	Senior, Tenant	The actor selects a matched profile and chooses the messaging option. The system opens a secure chat session and routes messages through encrypted communication channels. The actor can exchange messages with the matched user while the system securely stores the conversation history.
Report-User	WEB-FR-008	Senior, Tenant	The actor identifies suspicious, inappropriate, or unsafe behavior and submits a report through the reporting interface. The system records the report details and forwards the information to the administrative team for investigation. The reported activity is logged for future review and dispute resolution.

User Case Name	Related Requirements ID	Actor (s)	Brief Description
Manage-Reviews	APP-FR-013	Senior, Tenant	The actor navigates to the reviews section of a user profile to read feedback left by others. The system retrieves and displays all reviews associated with the selected profile. The actor uses this information to evaluate the trustworthiness and suitability of a potential match before proceeding.
View-Notifications	MOB-FR-014	Senior, Tenant	The actor accesses the notification center to review alerts related to matches, messages, and agreement updates. The system displays all unread and previously received notifications and allows the actor to navigate to the associated content. The actor can review notification details and track recent platform activity.

3.5.3 Table: Sub-systems/modules 3: Agreement & Administration

User Case Name	Related Requirements ID	Actor (s)	Brief Description
Create-Agreement	WEB-FR-009	Senior, Tenant	The actor initiates the creation of a living agreement after identifying a suitable match. The system presents a workspace where housing responsibilities, chores, rental discounts, and living arrangements can be documented. The completed agreement is stored and made available to both parties.

User Case Name	Related Requirements ID	Actor (s)	Brief Description
Modify/ Personalize- Agreement	WEB-FR-009	Senior, Tenant	The actor accesses an existing living agreement to review or update its contents. The system displays the agreement details and allows both parties to manage responsibilities and expectations. Any approved changes are saved and reflected in the latest version of the agreement.
Monitor-System	ADM-FR-012	System Administrator	The administrator accesses the management dashboard to monitor platform operations and activity. The system displays financial metrics, disputes, transaction information, and other operational indicators. The administrator uses this information to oversee system performance and identify potential issues.
Generate- Analytics	ADM-FR-012	System Administrator	The administrator requests analytical reports and key performance indicators from the dashboard. The system retrieves financial and operational data and presents the results in charts, summaries, and performance metrics. These analytics support business monitoring and strategic decision-making.

3.5.4 Table: Sub-systems/modules 4: Abstract Use Cases

User Case Name	Related Requirements ID	Actor (s)	Brief Description
Log-In	WEB-FR-001	N/A	The actor enters their username, email address, and password through the login interface. The system validates the submitted credentials and authenticates the user. Upon successful authentication, the system grants access to the platform and displays the user's dashboard.
Log-Out	WEB-FR-001	N/A	The actor selects the logout option from the platform interface. The system securely terminates the active session and removes authentication tokens. The actor is then redirected to the login or home page.
Reset-Password	WEB-FR-001	N/A	The actor selects the password recovery option after forgetting their login credentials. The system verifies the user's identity and sends a password reset link or verification code to the registered email address. The actor creates a new password, and the system updates the account credentials to restore access.

4. System Features

<This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product. You may prefer to organize this section by

use case, mode of operation, user class, object class, functional hierarchy, or combinations of these, whatever makes the most logical sense for your product.>

4.1 System Feature 1

<Don't really say "System Feature 1." State the feature name in just a few words.>

4.1.1 Description and Priority

<Provide a short description of the feature and indicate whether it is of High, Medium, or Low priority. You could also include specific priority component ratings, such as benefit, penalty, cost, and risk (each rated on a relative scale from a low of 1 to a high of 9).>

4.1.2 Stimulus/Response Sequences

<List the sequences of user actions and system responses that stimulate the behavior defined for this feature. These will correspond to the dialog elements associated with use cases.>

4.1.3 Functional Requirements List with Interview Traceability

Requirement ID	Requirement Title	Short Description	Priority	Requester
WEB-FR-001	Profile Identity Lifecycle & Onboarding	The system shall provide an onboarding setup broken into small, clear steps with simple language.	High	Margaret Liu Senior Homeowner
APP-FR-002	Document Upload	The system shall allow users to upload legal identification and photos to process identity verification.	High	Margaret Liu Senior Homeowner Jason Nguyen Tenant

Requirement ID	Requirement Title	Short Description	Priority	Requester
				Daniel Park System Administrator
APP-FR-003	Biometric Pass/Fail Gate	The system shall enforce identity validation during onboarding to confirm that every user's identity has been verified before they can interact with others on the platform.	High	Margaret Liu Senior Homeowner Jason Nguyen Tenant Daniel Park System Administrator
APP-FR-004	Automated Background Check Trigger	The system shall automatically trigger a background check for users once initial matching milestones are passed, to save costs and ensure both parties are screened before connecting.	High	Margaret Liu Senior Homeowner Jason Nguyen Tenant David Okonkwo Investor
APP-FR-005	Geospatial Discovery & Radius Mapping	The system shall allow users to search for listings by distance from a specified location such as a school or transit stop.	High	Jason Nguyen Tenant

Requirement ID	Requirement Title	Short Description	Priority	Requester
APP-FR-006	Algorithmic Match Score Evaluation	The system shall generate a compatibility score for each potential match, summarizing how well the two users' lifestyles and daily habits align.	High	Margaret Liu Senior Homeowner Jason Nguyen Tenant
APP-FR-007	WebSocket Chat Routing	The system shall provide in-app messaging that keeps all communication between users inside the platform without sharing personal contact information.	High	Margaret Liu Senior Homeowner Jason Nguyen Tenant
WEB-FR-008	In-App Reporting System	The system shall provide a way for users to report suspicious or unsafe behavior directly within the platform.	High	Margaret Liu Senior Homeowner Jason Nguyen Tenant
WEB-FR-009	Chore & Living Agreement Formulator	The system shall provide an agreement form written in plain language where users can outline household responsibilities and arrangements	High	Margaret Liu Senior Homeowner Jason Nguyen Tenant

Requirement ID	Requirement Title	Short Description	Priority	Requester
		before a tenant moves in.		
APP-FR-010	Headless CMS Documentation Integration	The system should provide a Digital Knowledge Base and Admin Manual that can be updated without code changes to support onboarding guidance and administrative documentation.	Medium	Sarah Chen Project Manager Daniel Park System Administrator
ADM-FR-011	Administrative Validation Overrides	The system's admin portal shall allow administrators to manually review flagged profiles, clear incorrect restrictions, and generate a log entry for each override action.	High	Daniel Park System Administrator
ADM-FR-012	Executive KPI Dashboard Analytics	The system's admin portal should display live financial metrics and platform activity so administrators and investors can monitor performance in real time.	Medium	David Okonkwo Investor Daniel Park System Administrator

Requirement ID	Requirement Title	Short Description	Priority	Requester
APP-FR-013	Manage Reviews	The system shall provide a reviews and feedback collection system where users can read reviews left by others.	High	Margaret Liu Senior Homeowner Jason Nguyen Tenant
MOB-FR-014	Automated Push Notification Engine	The system should send real-time notifications to users for new matches and messages.	Medium	Jason Nguyen Tenant

4.2 System Feature 2 (and so on)

4.3 High Priority Use Case Description & Swimlane Diagram

Use Case: Evaluate Compatibility

Iteration: 1. Last modification: n/a

Primary actor: Tenant, Senior

Goal in context: To generate a compatibility score between a Senior and a Tenant based on lifestyle variables to inform living arrangement decisions.

Preconditions: The user is logged into the **MatchMyHome** platform; the user is viewing a candidate profile on the discovery board or dashboard.

Trigger: The user selects a candidate profile from the discovery board or dashboard.

Scenario:

1. The User selects a candidate profile from the discovery board or dashboard.

2. The Match class initiates an asynchronous background job to process the compatibility evaluation.
3. The Tenant and Senior classes retrieve lifestyle preference data from both users from the MongoDB Atlas database.
4. The Match class filters out any banned discriminatory attributes to remain compliant with Provincial Human Rights Codes.
5. The Match class performs multi-variable comparison calculations.
6. The CompatibilityScore class caches the results in the Redis layer.
7. The CompatibilityScore class renders the compatibility percentage and highlights aligned lifestyle preferences on the Matching Matrix Panel.
8. The Conversation class enables the **Send-Message** functional option on the UI.

Exceptions:

1. The compatibility evaluation fails due to system error.
 - 1.1 The Match class displays an error message indicating that the compatibility score could not be generated at this time.
 - 1.2 The SystemMetrics class logs the error for administrative review.
2. The Match class exceeds the 150ms read latency threshold.
 - 2.1 The Match class displays a loading indicator until the calculations is complete and cached.

Priority: High. Implementation is essential.

When available: First increment.

Frequency of use: Frequent (per candidate profile view).

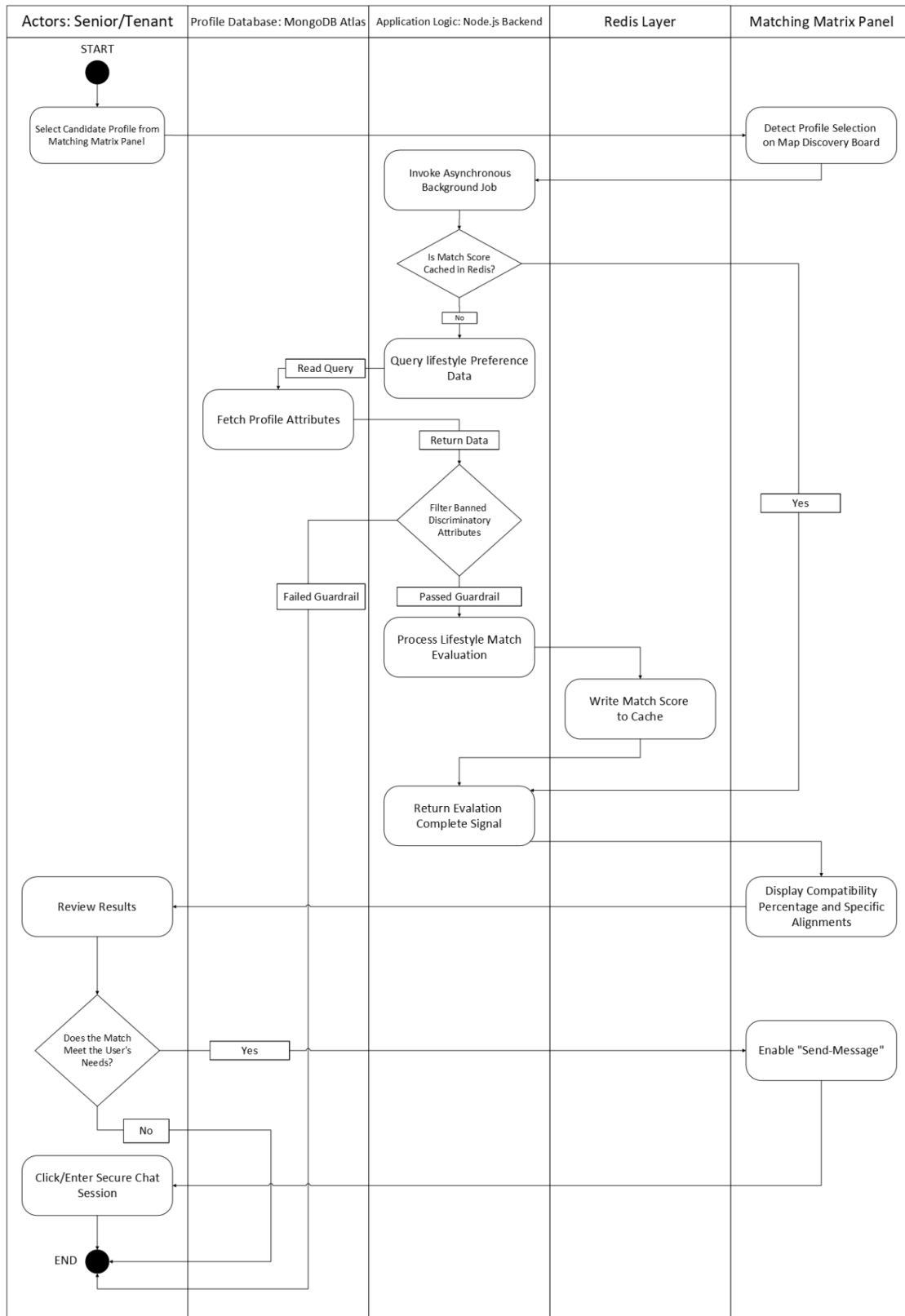
Channel to actor: Web browser; network connection.

Secondary actors: [Node.js](#) Backend, MongoDB Atlas, Redis, MatchMyHome Interface.

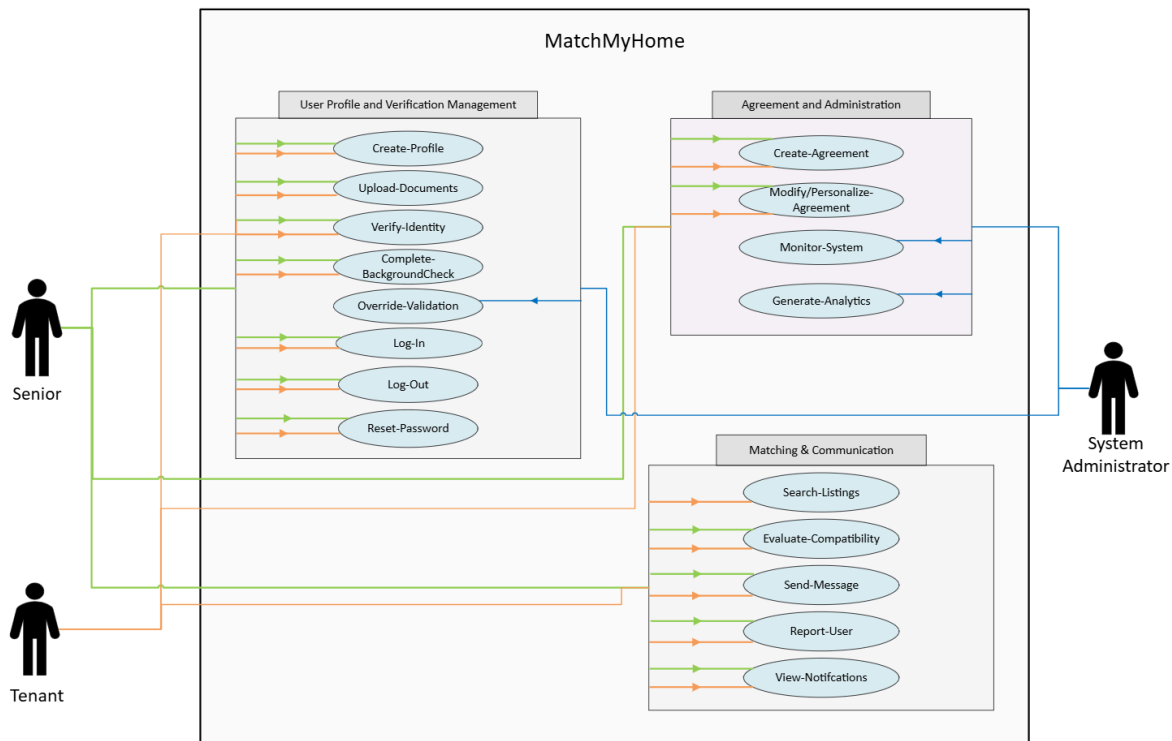
Channels to secondary actors:

- **[Node.js](#) Backend:** MatchMyHome interface; network connection.
- **MongoDB Atlas:** [Node.js](#) backend; database connection.
- **Redis:** [Node.js](#) backend; memory cache connection.
- **MatchMyHome Interface:** Web browser; network connection.

Swimlane Diagram



4.4 Use Case Diagram



5. Other Nonfunctional Requirements

5.1 Performance Requirements

<If there are performance requirements for the product under various circumstances, state them here and explain their rationale, to help the developers understand the intent and make suitable design choices. Specify the timing relationships for real time systems. Make such requirements as specific as possible. You may need to state performance requirements for individual functional requirements or features.>

5.2 Safety Requirements

<Specify those requirements that are concerned with possible loss, damage, or harm that could result from the use of the product. Define any safeguards or actions that must be taken, as well as

actions that must be prevented. Refer to any external policies or regulations that state safety issues that affect the product's design or use. Define any safety certifications that must be satisfied.>

5.3 Security Requirements

<Specify any requirements regarding security or privacy issues surrounding use of the product or protection of the data used or created by the product. Define any user identity authentication requirements. Refer to any external policies or regulations containing security issues that affect the product. Define any security or privacy certifications that must be satisfied.>

5.4 Software Quality Attributes

<Specify any additional quality characteristics for the product that will be important to either the customers or the developers. Some to consider are: adaptability, availability, correctness, flexibility, interoperability, maintainability, portability, reliability, reusability, robustness, testability, and usability. Write these to be specific, quantitative, and verifiable when possible. At the least, clarify the relative preferences for various attributes, such as ease of use over ease of learning.>

5.5 Business Rules

<List any operating principles about the product, such as which individuals or roles can perform which functions under specific circumstances. These are not functional requirements in themselves, but they may imply certain functional requirements to enforce the rules.>

5.6 Non-Functional Requirements List with Interview Traceability

Requirement ID	Requirement Title	Short Description	Priority	Requester
APP-NFR-001	Read/Write Database Latency	MongoDB Atlas and Redis caching shall maintain a read latency of < 15 ms (p95) for cached data and < 150 ms (p99) for live queries.	High	Michael Rodriguez Developer
APP-NFR-002	High-Traffic Throughput Capacity	The system's background matching routines shall evaluate a minimum of 1,000,000 multi-variable profile comparisons per 10 minutes, supporting up to 10,000 concurrent sessions and maintaining a throughput scale of $\geq 2,500$ operations per second under heavy traffic.	High	Michael Rodriguez Developer
APP-NFR-003	Server Resource Optimization	Under peak platform utilization or matching routine execution, the system's central database CPU utilization should keep $\leq 70\%$.	Medium	Michael Rodriguez Developer
WEB-NFR-004	WCAG 2.1 AA Accessibility Standards	The system's web platforms shall support text views scaling up to 200%, a minimum 4.5:1 contrast ratio, high-contrast toggles, and clear keyboard tab sequences.	High	Michael Rodriguez Developer

Requirement ID	Requirement Title	Short Description	Priority	Requester
APP-NFR-005	Legal & Privacy Compliance Testing	The system's test suite shall run automated integration verification scripts confirming sensitive data is encrypted and permanently auto deleted per PIPEDA.	High	Daniel Park System Administrator David Okonkwo Investor
APP-NFR-006	Algorithmic non-discrimination	The system's matching logic shall actively block, or strip out banned discriminatory filters (race, religion, orientation) per Provincial Human Rights Codes.	High	David Okonkwo Investor Jason Nguyen Tenant
WEB-NFR-007	Automated Cross-Browser Compatibility	The system's testing pipelines should execute automated UI tests across desktop browsers (Chrome v110+, Firefox v110+, Safari v16+) and mobile web browsers (Safari Mobile and Chrome for Android) to verify layout and onboarding stability.	Medium	Michael Rodriguez Developer
APP-NFR-008	Automated System Brute-Force Alerting	The system shall continuously monitor authentication traffic and run automated real-time alerting systems to instantly flag and block brute-force attack vectors.	High	Daniel Park System Administrator

6. Other Requirements

<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: Analysis Models

<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>

Appendix C: Stakeholder Registry

Table C.1: MatchMyHome Stakeholder Registry

Stakeholder Name	Stakeholder Position	Internal / External	Contact Details	Operational / Executive	Interest Level
Sarah Chen	Project Manager	Internal	sarah.chen@matchmyhome.ca	Operational	High
Michael Rodriguez	Developer	Internal	michael.rodriguez@matchmyhome.ca	Operational	High

Stakeholder Name	Stakeholder Position	Internal / External	Contact Details	Operational / Executive	Interest Level
Daniel Park	System Administrator	Internal	daniel.park@matchmyhome.ca	Operational	High
Margaret Liu	Senior Homeowner (Host)	External	margaret.liu@gmail.com	Operational	High
Jason Nguyen	Tenant	External	jason.nguyen@gmail.com	Operational	High
David Okonkwo	Investor	External	david.okonkwo@venturegroup.ca	Executive	High

Primary User Persona - Host Profile

Eleanor Vance

Demographics & Profile

- **Age:** 72
- **Location:** Victoria, British Columbia, Canada
- **Occupation:** Retired High School Literature Teacher (Widowed)
- **Living Situation:** Owns a spacious, fully paid-off 4-bedroom suburban home that feels increasingly empty and costly to maintain.
- **Tech-Savviness:** Moderate. Comfortable with email, online banking, and text-based communication, but requires clean layouts with high-contrast text and accessible design elements due to physical adaptations.
- **Physical Considerations:**

1. **Hearing Impairment:** Wears a digital behind-the-ear hearing aid, meaning phone and platform notifications require visual flashing or vibrating cues (synced via external desk puck or accessibility hardware).
2. **Hand Dexterity:** Suffers from mild arthritis, resulting in limited mobility and stiffness in her fingers, making small touch targets, tight typing spaces, or heavy swiping gestures difficult to navigate. Therefore, she relies heavily on voice dictation and large, simplified interactive elements rather than traditional typing.

Appendix D: Interview Questions

Table D.1: **Lead Investor** Interview Questions and Answers

Question	Stakeholder Position	Answer
1. What are the financial and operational milestones required prior to regional expansion?	Investor	Expansion Thresholds: Achieve ≥ 500 active, verified matches maintained for 6 consecutive months across pilot cities (Toronto, Vancouver), maintain Host churn rate $< 5\%$, and verify platform transaction margins cover operational costs.
2. What are the target launch metrics for user acquisition, activation, and platform safety?	Investor	Launch Targets: Maintain a 3:1 Tenant-to-Host ratio. Achieve an 80% Host onboarding-to-listing conversion rate within 14 days. Maintain an Incident-Free Match Rate $\geq 99.8\%$ (fewer than 2 critical incidents per

Question	Stakeholder Position	Answer
		1,000 active matches reported to System Admins).
3. What is the deployment schedule for premium features relative to the MVP launch?	Investor	Phased Rollout: Identity verification (Certn/Sterling) and secure payments (Stripe) are mandatory MVP launch requirements. The Chore & Living Agreement Formulator will deploy as a static template in Phase 1 and upgrade to a dynamic tool in Phase 2.
4. What core financial metrics must the executive KPI dashboard track via the Stripe API?	Investor	Dashboard Analytics: The system must track Gross Merchandise Value (GMV), net platform revenue, chore-based rental discount values, and billing/refund disputes handled by System Administrators.
5. How should the system minimize third-party API transaction costs for background checks?	Investor	Workflow Optimization: The system must restrict Certn/Sterling background screening triggers to users who have successfully passed initial algorithmic matching milestones, preventing costs on idle profiles.

Table D.2: **Project Manager** Interview Questions and Answers

Question	Stakeholder Position	Answer
1. How should development be prioritized between the Web and Mobile (React Native) platforms?	Project Manager	Platform Prioritization: Phase 1 (MVP) will prioritize a responsive React Web Application optimized for senior accessibility (desktop/tablet layouts). Standalone iOS/Android apps are deferred to Phase 2.
2. What architectural safeguards are required to handle third-party API downtime?	Project Manager	Fault Tolerance: The Node.js middleware must feature robust offline error-handling capabilities, circuit-breaker patterns, and localized data caching to ensure external API outages do not crash the UI.
3. What is the integration strategy for the backend screening pipelines and frontend onboarding?	Project Manager	Development Methodology: Frontend (React wizard) and backend (Node.js/MongoDB) workflows must run in parallel using mocked API payloads. Final system integration is scheduled for a dedicated Week 6 sprint.
4. How will the platform manage and update user onboarding and administrative support documentation?	Project Manager	Content Management: The Senior Onboarding wizard must be hardcoded into the frontend. The Digital Knowledge Base and Admin Manual must utilize a headless CMS (e.g., Contentful) for decoupled, code-free updates.

Question	Stakeholder Position	Answer
5. What is the engineering protocol if a data compliance or housing law violation is identified?	Project Manager	Compliance Escalation: Any identified PIPEDA or provincial housing law violation mandates an immediate sprint freeze, emergency backlog grooming, and complete reallocation of resources to schema remediation.

Table D.3: **Developer** Interview Questions and Answers

Question	Stakeholder Position	Answer
1. How will the MongoDB Atlas schema be optimized to calculate the Lifestyle Match Score efficiently?	Developer	Fast Match Calculation: Store lifestyle preferences as flat data inside the user profile. Use indexed database queries combined with a Redis caching layer to achieve a read latency of < 15 ms (p95) for cached data and < 150 ms (p99) for live queries . Complex multi-variable match scores will be calculated via asynchronous background jobs capable of processing 1,000,000 evaluations per 10 minutes , keeping the user interface lightning-fast ($\geq 2,500$ operations per second) during heavy traffic up to a peak load of 10,000 concurrent sessions, while keeping database CPU utilization $\leq 70\%$.

Question	Stakeholder Position	Answer
2. What technical accessibility standards should the web platform meet to support senior users?	Developer	Accessibility Standards: The web platform shall support text scaling up to 200% without loss of content, maintain a minimum contrast ratio of 4.5:1 for all text elements, provide high-contrast display toggles, and ensure all interactive elements are reachable via clear keyboard tab sequences to comply with WCAG 2.1 AA.
3. How will real-time messaging be implemented to protect user privacy and metadata?	Developer	Private Messaging: Route chat traffic through secure WebSockets (<code>wss://</code>). The server will act as a middleman to strip away hidden metadata, verify user identities, and encrypt chat histories before saving them.
4. How will radius-based geospatial searches be handled on the Node.js backend?	Developer	Location Searches: Store property coordinates using MongoDB's native location format (GeoJSON). Run radius queries directly in the database using a <code>2dsphere</code> index, then send the clean coordinates to Google Maps to draw the search area.

Question	Stakeholder Position	Answer
5. What database schema structure will guarantee an immutable audit ledger for user reviews?	Developer	Tamper-Proof Ledger: Set the reviews collection to Append-Only. Database rules will block all update and delete requests from users. Any corrections must be added as a brand-new entry, giving admins an unchangeable history trail.

Table D.4: **System Administrator** Interview Questions and Answers

Question	Stakeholder Position	Answer
1. How do you currently handle situations where the automated identity verification system flags a user profile incorrectly?	System Administrator	Manual Override Protocol: When the automated validation system flags a profile incorrectly, I access the Admin Control Center and review the submitted documents manually. I compare the flagged fields against the original uploads and, if the profile is legitimate, I use the administrative override function to clear the restriction. A log entry is automatically generated for audit purposes.

Question	Stakeholder Position	Answer
2. What information do you need visible on the admin dashboard to monitor platform health effectively on a daily basis?	System Administrator	Dashboard Requirements: The dashboard must display real-time counts of active users, pending verifications, flagged profiles, and open disputes. I also need visibility into match success rates, background check statuses from Certn/Sterling, and any brute-force or suspicious login alerts so I can respond to security incidents without delay.
3. How should the platform handle a reported user while an investigation is still ongoing?	System Administrator	Suspension Workflow: When a report is submitted, the reported user's ability to send new messages and initiate new matches should be temporarily suspended pending review. I need the ability to escalate the case, add internal notes, and either reinstate or permanently ban the account once the investigation concludes. The reporting user should receive an automated acknowledgment.

Question	Stakeholder Position	Answer
4. What level of access should the System Administrator have over active living agreements between users?	System Administrator	Agreement Oversight: I should be able to view all active agreements stored in the Contract Ledger Board for dispute resolution and compliance purposes, but I should not be able to modify the content of an agreement without a logged justification. Edits made by administrators must be flagged and visible in the audit trail to maintain platform integrity.
5. How should the platform notify you when a critical system alert or security threshold is breached?	System Administrator	Alert Notification: Critical alerts — such as a brute-force login threshold being exceeded or a background check API failure — should trigger an immediate push notification to my admin account and generate a visible banner on the Admin Control Center dashboard. Alerts must include a timestamp, affected user ID or system component, and a recommended action step.

Table D.5: **Senior Homeowner** Interview Questions and Answers

Question	Stakeholder Position	Answer
1. What concerns do you have about sharing your home with a stranger, and	Senior Homeowner	Safety First: My biggest concern is not knowing who is coming into my home. Before I would consider using

Question	Stakeholder Position	Answer
how would a platform need to address them before you would feel comfortable signing up?		the platform, I would need to know that every applicant has passed a proper background check and that their identity has been verified with real documents. I would also want to read reviews from other homeowners who have hosted them before I agree to anything.
2. How comfortable are you using technology, and what kind of help would you need to set up your profile and listing?	Senior Homeowner	Guided Onboarding: I am comfortable with basic computer and tablet use, but filling out complex forms online can be confusing. I would need the profile setup to be broken into small, clear steps with simple language and large text. If something goes wrong, I would want a help button nearby or a way to call someone for assistance rather than having to figure it out on my own.
3. What information about a potential tenant is most important to you when deciding whether to invite them for a meeting?	Senior Homeowner	Tenant Evaluation: I want to see a photo, their age, why they are looking for a home-sharing arrangement, and what kind of daily schedule they keep. Quiet hours, cleanliness habits, and whether they smoke or have pets matter a lot to me. A compatibility score that summarizes how well our

Question	Stakeholder Position	Answer
		lifestyles align would help me make a faster and more confident decision.
4. How would you prefer to communicate with a potential tenant before agreeing to a home-sharing arrangement?	Senior Homeowner	Communication Preferences: I would prefer to message through the app first to get a sense of the person, and then move to a video call or an in-person meeting before making any decision. I would not want my phone number or personal email shared with anyone through the platform. All communication should stay inside the app until I am ready to take the next step.
5. What household responsibilities or arrangements would you want clearly written down before a tenant moves in?	Senior Homeowner	Agreement Expectations: I would want an agreement that lists which chores the tenant is responsible for, such as helping with groceries or light yard work, and what I am offering in return, such as a reduced rent. I would need the workspace tool to guide us using plain language while the system safely generates a legally compliant framework agreement underneath. Having a copy I can print or save on my phone would also be important to me.

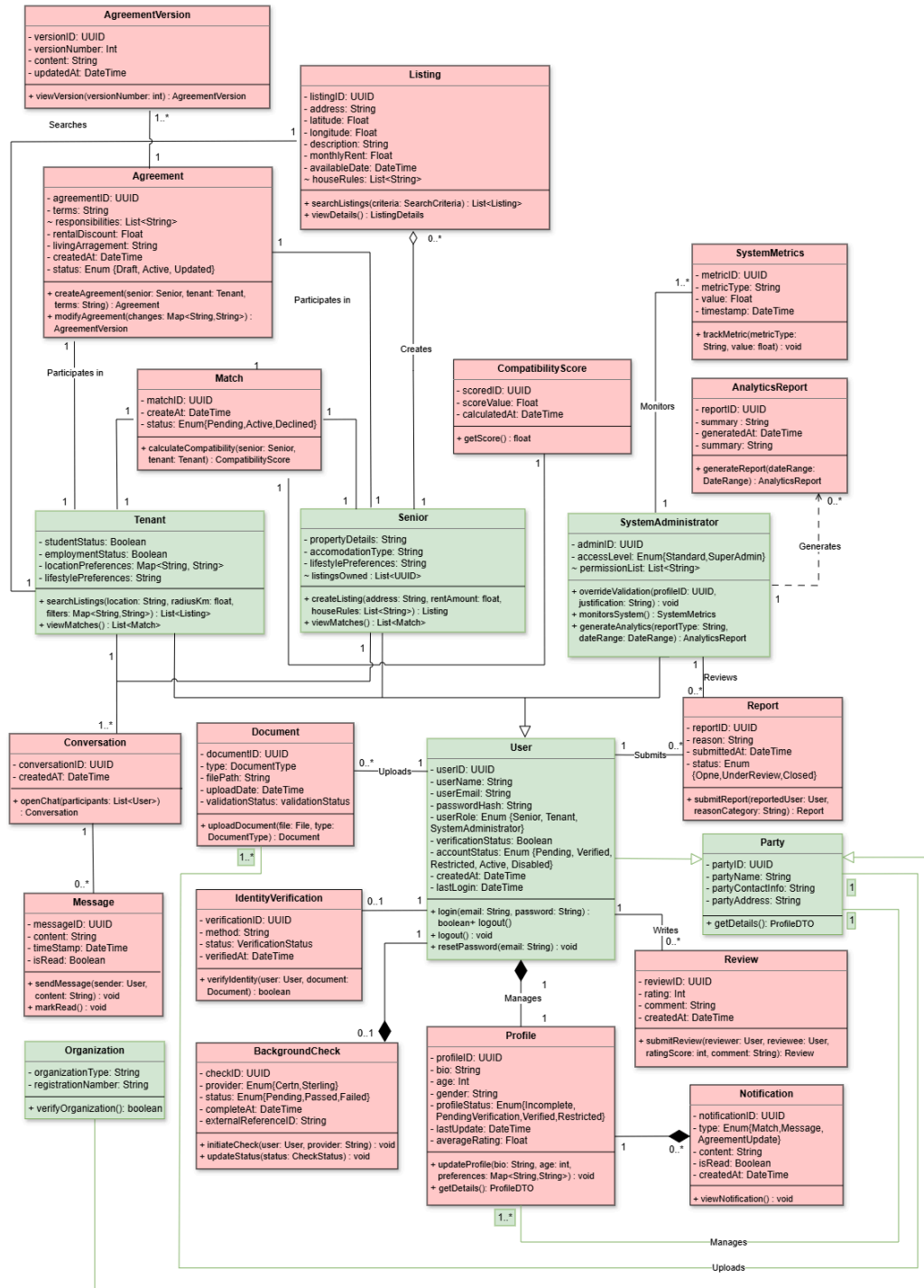
Table D.6: **Tenant** Interview Questions and Answers

Question	Stakeholder Position	Answer
1. What are the most important factors you consider when searching for affordable housing, and how does your current situation make that difficult?	Tenant	Housing Priorities: Affordability is the biggest factor, followed by proximity to my school or workplace. Right now, the rental market is very competitive and most listings I find are either too expensive or already taken by the time I apply. I am also cautious about safety and would want to know about the neighbourhood and the homeowner before committing to anything.
2. How would you feel about living with a senior homeowner, and what would make that arrangement appealing or unappealing to you?	Tenant	Co-Living Perspective: I think it could be a great arrangement if the expectations are clear from the start. The idea of lower rent in exchange for some help around the house makes practical sense for me. What would make it unappealing is if there were no structure around what I am expected to do or if the homeowner's lifestyle was very different from mine with no compatibility check beforehand.
3. What information would you want to see in	Tenant	Listing Requirements: I would like to see photos of the room and common

Question	Stakeholder Position	Answer
a homeowner's listing before deciding whether to apply?		areas, the monthly cost or rent-reduction arrangement, the location on a map, house rules such as quiet hours or guest policies, and a brief profile of the homeowner. A compatibility summary showing how well our habits align, such as sleep schedules and cleanliness, would also help me decide quickly.
4. How would you prefer to search for listings, and what filters would be most useful to you?	Tenant	Search Preferences: I would want to search by distance from my school or transit stop, monthly cost, and room size. Filters for pet-friendly homes, available move-in dates, and required chore commitments per week would also be very helpful. Being able to save listings and come back to them later without losing my progress would make the experience much smoother.
5. What would make you feel safe enough to move forward with a home-sharing arrangement through a platform you have never used before?	Tenant	Trust Requirements: Knowing that the homeowner has been identity-verified and that background checks have been run on both sides would give me confidence. Seeing real reviews from previous tenants would also matter. I would also want the agreement terms

Question	Stakeholder Position	Answer
		to be written clearly in the app so there is no confusion about what I agreed to, and I would need a way to report concerns if something felt unsafe after moving in.

Class Diagram



The Core Concept (Fowler's Party Pattern)

In enterprise and business domain modeling, software frequently needs to track entities that participate in transactions, agreements, or listings - whether they are single human beings or corporate bodies.

Martin Fowler introduced the **Party** pattern as a generic supertype to represent *any* general actor or legal entity. Instead of creating separate, unrelated structures for different types of actors, the pattern establishes a single root abstraction (Party) that can be specialized into concrete subtypes like a **User** (an individual person) or an **Organization** (a company or corporate landlord).

How the Classes Relate

Party (Supertype): Holds all the shared, universal attributes that apply to any participant in our **MatchMyHome** application - such as a unique identifier (partyID), names, contact info, and addresses.

User & Organization (Subtypes): Both inherit from Party via **Generalization** (the hollow triangle inheritance arrow).

The user keeps individual-specific attributes (like login credentials, passwords, and personal session data).

Organization handles corporate-specific attributes and behaviors (like registrationNumber and verifyOrganization()).

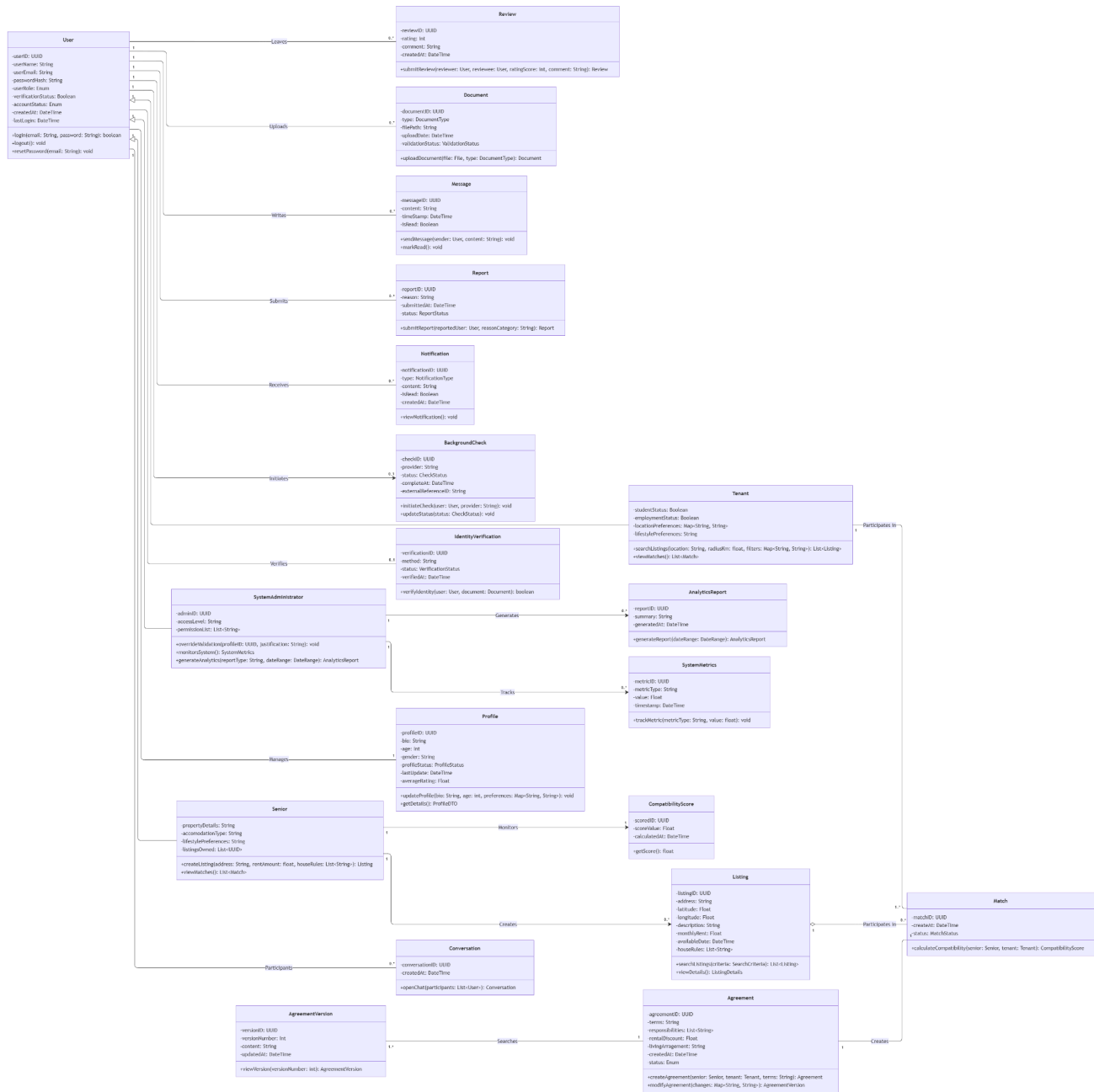
Why This Approach is Valid and Effective

Prevents Code and Data Duplication: Instead of forcing corporate data fields into a human user profile or vice versa, shared traits are centralized once in Party.

Open for Extension: If the platform scales later to include government bodies, insurance partners, or cleaning vendors, we can add them effortlessly as new subtypes of Party without breaking existing system architecture.

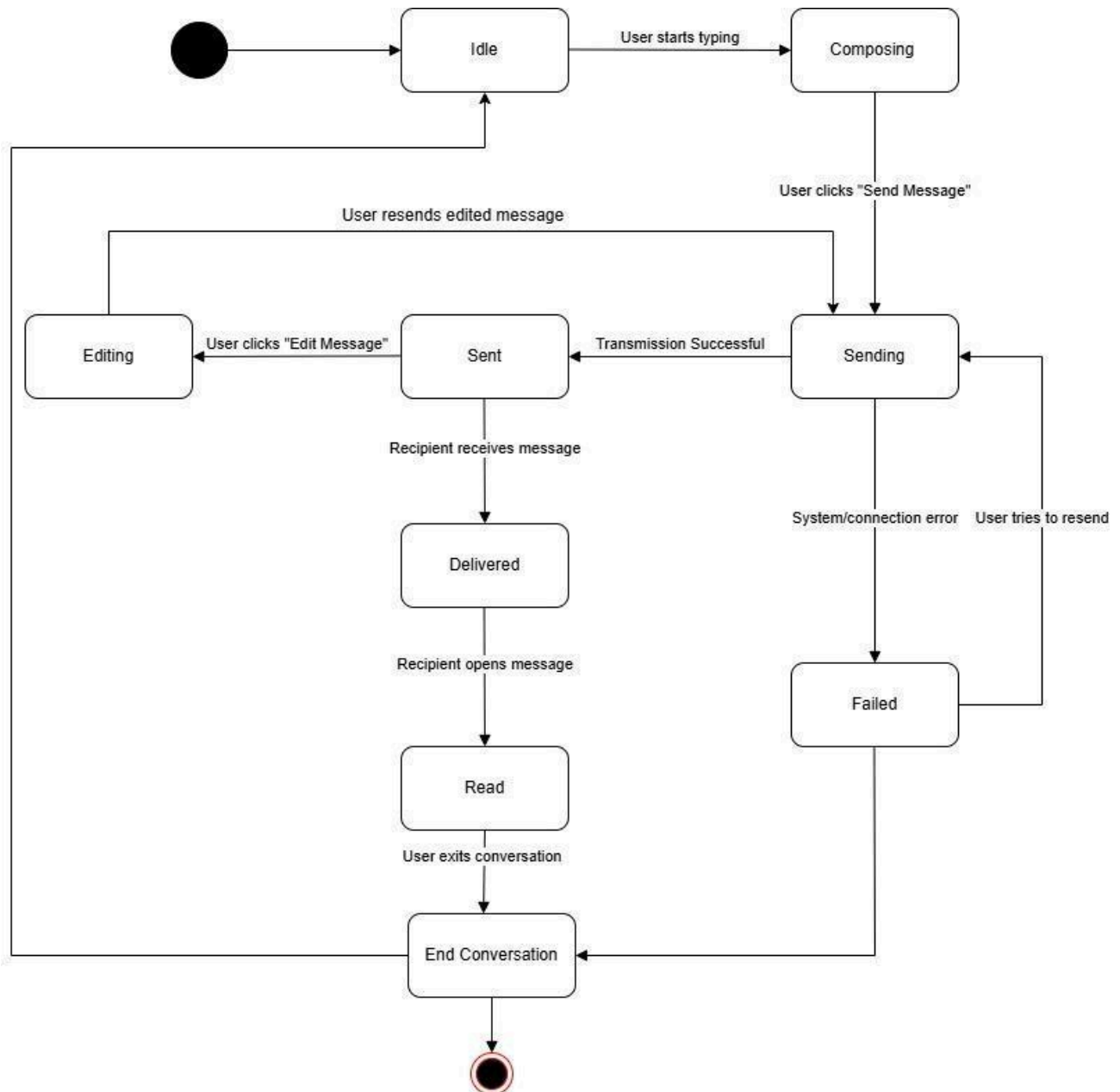
- **Meets Academic Standards:** It demonstrates professional proficiency in object-oriented design principles (inheritance, abstraction, and modularity) required for structural architecture documentation.

Gen AI Class Diagram

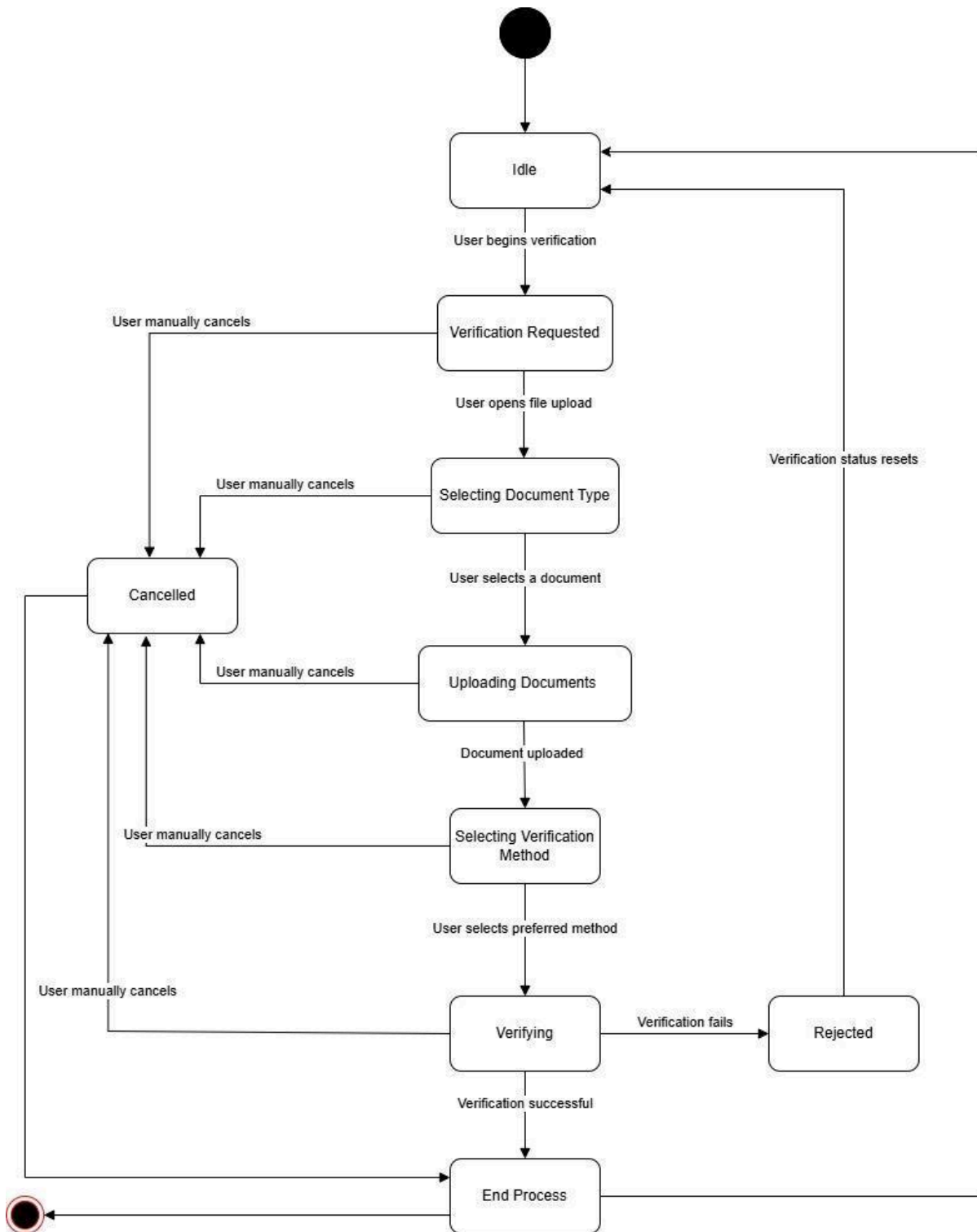


State Diagrams

Message Class State Diagram

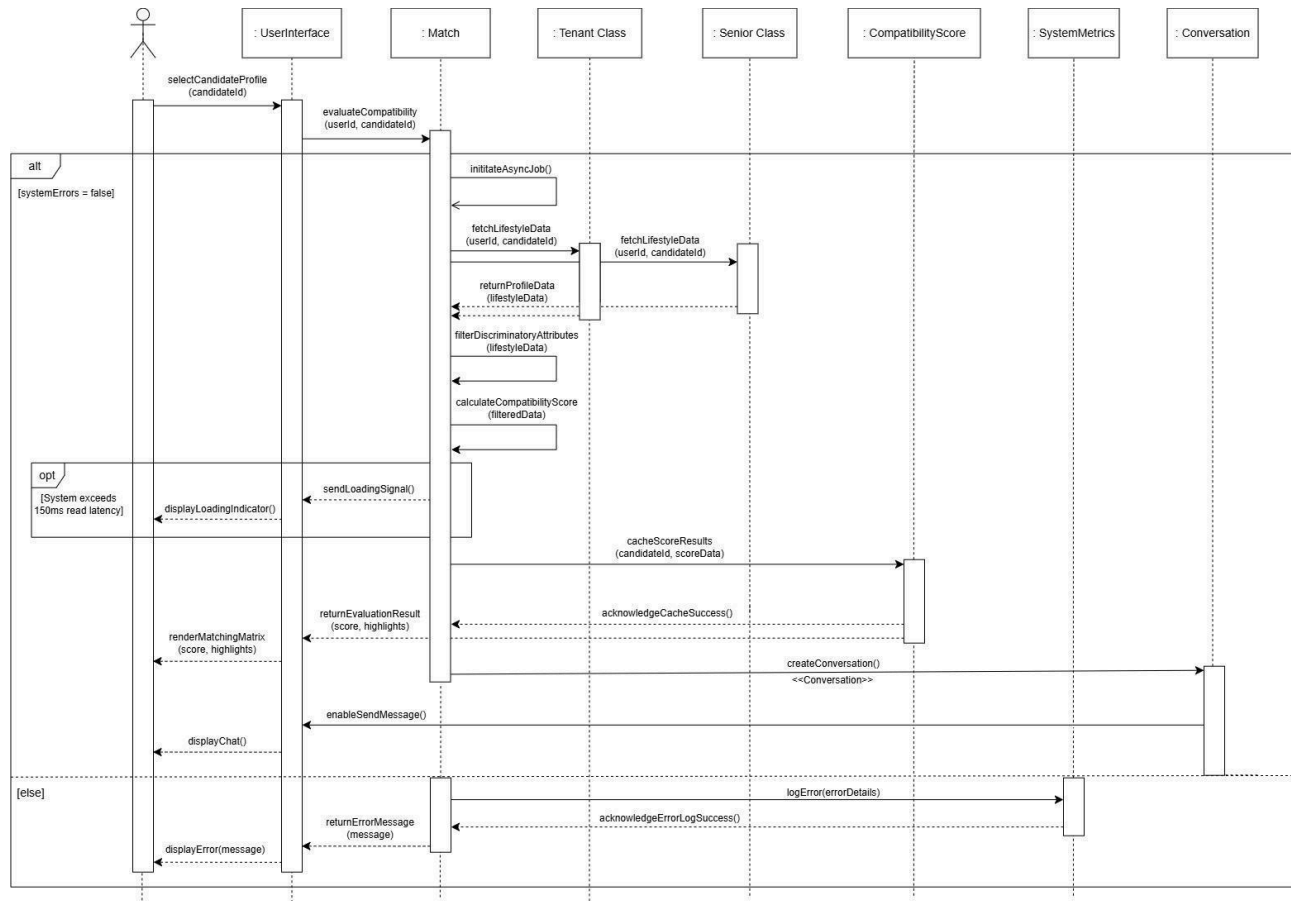


IdentityVerification Class State Diagram

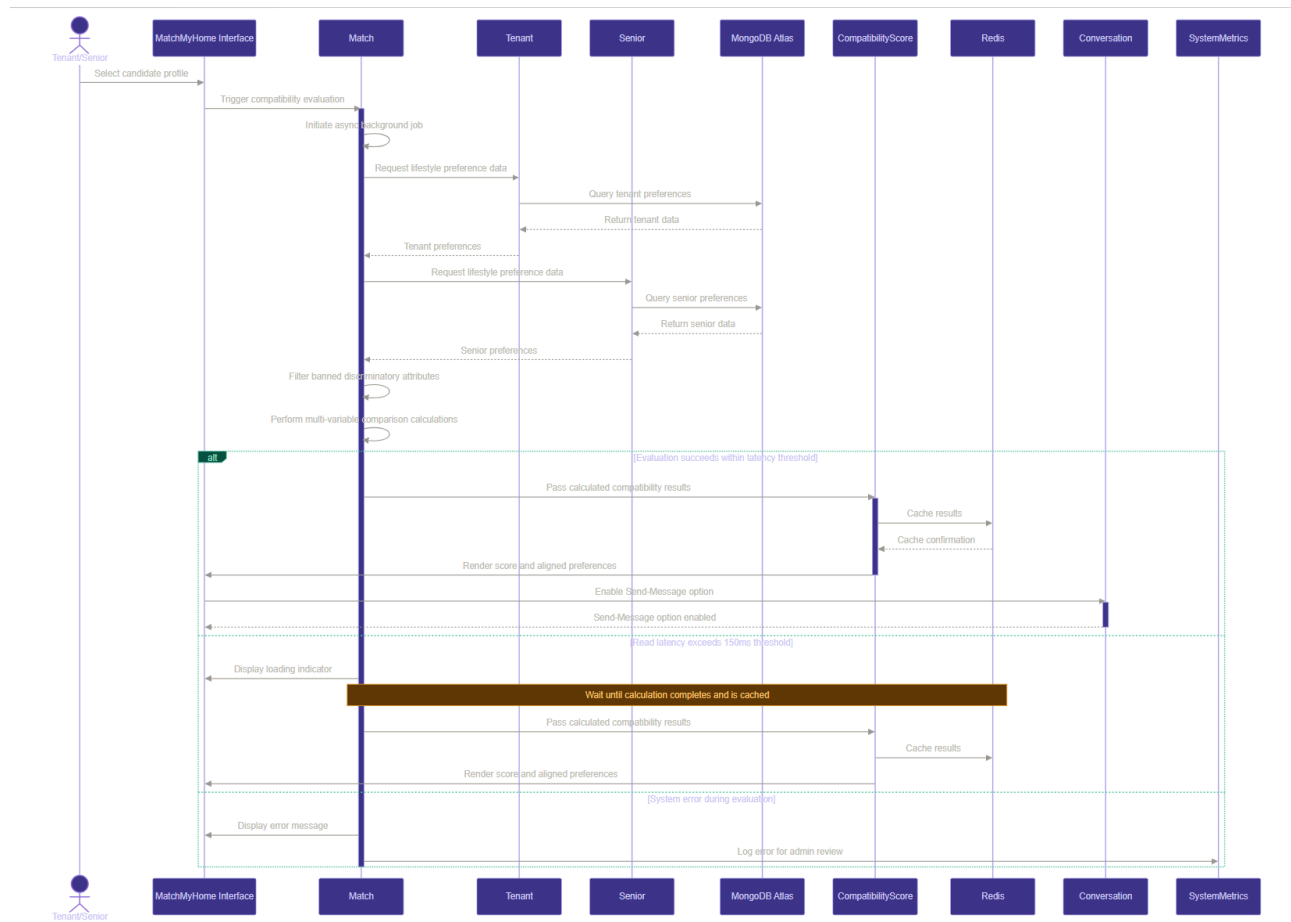


Sequence Diagram

Use Case: Evaluate Compatibility



Gen AI Sequence Diagram



CRC Cards

Class: User

Class Name: User		Superclass: N/A Subclasses: Senior, Tenant, SystemAdministrator
<i>Description: Represents a platform user with core authentication, identity, and account lifecycle information</i>		
Attributes: - userID: UUID — Unique identifier for the user account. - email: String — Registered primary email address used for login and notifications - passwordHash: String — Secured salt-hashed credential value for authentication + accountStatus: Enum — Current operational lifecycle state (Active, Suspended, Pending)		
Responsibilities / Methods		Collaborators
+ login(IdentityVerification)		IdentityVerification
+ logout()		N/A
+ resetPassword()		N/A

Class: Profile

Class Name: Profile		Superclass: N/A Subclasses: N/A
<i>Description: Represents a user's public-facing profile, including personal details, status, and reputation indicators</i>		
Attributes: - profileID: UUID — Unique identifier for the public profile record + bio: Text — User-written introduction and descriptive background summary + age: int — Calculated numeric age of the individual profile actor + averageRating: float — Calculated mean rating from all completed peer reviews		
Responsibilities / Methods		Collaborators
+ updateProfile()		User
+ getDetails()		User

Class: Senior

Class Name: Senior		Superclass: User Subclasses: N/A
<i>Description: Represents a senior homeowner offering living space within their property</i>		
Attributes: - seniorID: int — Primary specific key identifier mapping to base user domain ~ listingsOwned: List<UUID> — Collection reference array tracking property postings owned		
Responsibilities / Methods		Collaborators
+ createListing()		Listing
+ viewMatches()		Match

Class: Tenant

Class Name: Tenant		Superclass: User Subclasses: N/A
<i>Description: Represents a student or young professional seeking housing accommodations</i>		
Attributes: - tenantID: int — Primary unique domain identifier extension for tenant users + institutionName: String — Verified name of college or university enrollment status		
Responsibilities / Methods		Collaborators
+ searchListings()		Listing
+ viewMatches()		Match

Class: Listing

Class Name: Listing		Superclass: N/A Subclasses: N/A
<i>Description: Represents a housing listing created by a senior containing location and property details</i>		
Attributes: - listingID: UUID — Unique persistent primary reference key identifier + rentAmount: float — The base localized monthly financial cost allocation + houseRules: List<String> — Explicit living conditions and habit guidelines		
Responsibilities / Methods		Collaborators
+ searchListings()		Tenant
+ viewDetails()		Tenant

Class: Match

Class Name: Match		Superclass: N/A Subclasses: N/A
<i>Description: Represents an algorithmically proposed pairing link between a tenant and a listing</i>		
Attributes: - matchID: UUID — Primary matching ledger transactional record tracking item key + matchStatus: Enum — Tracks step pipeline configuration status (Pending, Accepted, Declined)		
Responsibilities / Methods		Collaborators
+ calculateCompatibility()		CompatibilityScore

Class: CompatibilityScore

Class Name: CompatibilityScore		Superclass: N/A Subclasses: N/A
<i>Description: Represents the numeric compatibility score calculated between users or listings</i>		
Attributes: - scoreID: int — Unique identifier for the processed calculation record + overallPercentage: float — Aggregated final alignment match accuracy rating scale		
Responsibilities / Methods		Collaborators
+ getScore()		Match

Class: Conversation

Class Name: Conversation		Superclass: N/A Subclasses: N/A
<i>Description: Represents a conversational communication chat thread session block</i>		
Attributes: - conversationID: UUID — Primary persistent database channel identifier tracking key ~ isArchived: boolean — Flag tracking if chat history visibility is hidden		
Responsibilities / Methods		Collaborators
+ openChat()		User

Class: Message

Class Name: Message		Superclass: N/A Subclasses: N/A
<i>Description: Represents an individual discrete message node within an active conversation thread</i>		
Attributes: - messageID: UUID — Unique tracking key identifying individual textual entries + messageText: String — Raw text payload contents of transmitted data frame		
Responsibilities / Methods		Collaborators
+ sendMessage()		Conversation
+ markRead()		User

Class: Review

Class Name: Review		Superclass: N/A Subclasses: N/A
<i>Description: Persistent feedback records containing rating scores and verification feedback comments</i>		
Attributes: - reviewID: int — Unique tracking reference assigned upon feedback submission + ratingScore: int — Integer value ranking scale score (1 to 5 stars)		
Responsibilities / Methods		Collaborators
+ submitReview()		Senior, Tenant

Class: Document

Class Name: Document		Superclass: N/A Subclasses: N/A
<i>Description: Represents an uploaded file object payload used inside verification workflows</i>		
Attributes: - documentID: UUID — Unique primary storage tracking indexing value + fileUrl: String — Secure target content delivery link string locator pathway		
Responsibilities / Methods		Collaborators
+ uploadDocument()		User

Class: IdentityVerification

Class Name: IdentityVerification		Superclass: N/A Subclasses: N/A
<i>Description: Validates user identities using biometric data or official documentation</i>		
Attributes: - verificationID: UUID — Unique identifier for the tracking record validation cycle + status: Enum — Current verification operational check state output		
Responsibilities / Methods		Collaborators
+ verifyIdentity()		User, Document

Class: BackgroundCheck

Class Name: BackgroundCheck		Superclass: N/A Subclasses: N/A
<i>Description: Coordinates automated criminal tracking check pipelines using external APIs</i>		
Attributes: - checkID: UUID — Transactional unique screening identifier token block ID + status: Enum — External processing response codes mapping tracker indicator		
Responsibilities / Methods		Collaborators
+ initiateCheck()		User
+ updateStatus()		SystemAdministrator

Class: Report

Class Name: Report		Superclass: N/A Subclasses: N/A
<i>Description: Tracks user complaints or safety incident events</i>		
Attributes: - reportID: UUID — Unique identification index tracking the incident ticket item + reasonCategory: String — Subject grouping indicator describing the incident		
Responsibilities / Methods		Collaborators
+ submitReport()		User

Class: Agreement

Class Name: Agreement		Superclass: N/A Subclasses: N/A
<i>Description: The core operational contract detailing guidelines, rents, and chores</i>		
Attributes: - agreementID: UUID — Unique primary lease document configuration tracker token ID + choreList: List<String> — Assigned community service task mappings and frequencies		
Responsibilities / Methods		Collaborators
+ createAgreement()		AgreementVersion
+ modifyAgreement()		AgreementVersion

Class: AgreementVersion

Class Name: AgreementVersion Superclass: N/A Subclasses: N/A	
<i>Description: An immutable ledger captures logging version history checkpoints</i>	
Attributes: - versionID: UUID — Unique tracking iteration block identifier value ID + contentSnapshot: String — Complete structural backup copy of contract state records	
Responsibilities / Methods	Collaborators
+ viewVersion()	Agreement

Class: SystemAdministrator

Class Name: SystemAdministrator Superclass: User Subclasses: N/A	
<i>Description: Internal control tier role managing application maintenance and validation overrides</i>	
Attributes: - adminID: int — Unique administrator staff account grouping ID identification code ~ permissionsList: List<String> — System authorization control validation scope maps	
Responsibilities / Methods	Collaborators
+ overrideValidation()	IdentityVerification
+ monitorsSystem()	SystemMetrics
+ generateAnalytics()	AnalyticsReport

Class: SystemMetrics

Class Name: SystemMetrics		Superclass: N/A Subclasses: N/A
<i>Description: Tracks performance and traffic analytical signals continuously</i>		
Attributes: - metricID: int — Unique incremental metric tracking baseline frame key + systemLatencyMs: int — Current backend round-trip response metrics time indicator		
Responsibilities / Methods		Collaborators
+ trackMetric()		SystemAdministrator

Class: AnalyticsReport

Class Name: AnalyticsReport		Superclass: N/A Subclasses: N/A
<i>Description: Aggregates analytical logs into formal managerial reports</i>		
Attributes: - reportID: int — Primary persistent key code tracking generated outputs document + dataPayload: String — Raw performance summary serialization payload text map metadata		
Responsibilities / Methods		Collaborators
+ generateReport()		CompatibilityScore, SystemAdministrator

Class: Notification

Class Name: Notification		Superclass: N/A Subclasses: N/A
<i>Description: Dispatches transactional real-time informational status updates to users</i>		
Attributes: - notificationID: UUID — Unique tracing identification code element + isRead: boolean — Flag identifying if user has seen the alert notification banner		
Responsibilities / Methods		Collaborators
+ viewNotification()		User